Module code: MTHA5003A
Module name: Complex Functions & Linear Algebra
Deadline: Monday 28th October 2024, 15:00
Hand in to: via Blackboard

University of East Anglia
School of Mathematics

Coursework 2 - Linear Algebra

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General Instructions

Attempt all questions. Your work will be marked according to the following descriptors. In particular, please note that a solution which is numerically correct but contains no explanatory words will not score more than 60%, and solutions should be written in complete English sentences to get a mark of 80% or above.

91-100% - A clear and correct solution which shows insight or innovation beyond what is needed, but not at the cost of making the solution to the original problem unclear.

81-90% - A completely clear and correct solution with all key ideas present and no extraneous information.

71-80% - A correct solution which is clear apart from a small number of minor issues, or a very clear solution which is correct apart from minor errors.

61-70% - A correct solution which is mostly clear, or a clear solution which contains most of the key ideas and is almost correct.

51-60% - An essentially correct solution which is not clearly written or explained, or a partial solution containing some key ideas clearly explained.

41-50% - Most of a full solution but not clearly explained, or some relevant ideas explained.

31-40% - Relevant ideas present and understandable, although perhaps not very clearly explained.

1-30% - Some fragmentary ideas relevant to the question present, but not close to a full solution.

0% - No reasonable attempt made.

Coursework 2 - Linear Algebra

1. Let V be a vector space of dimension n. Let $T : V \to V$ be a linear map such that $T^2 = 0$ (i.e. T composed with itself is the zero map).

a. Show that $\mathrm{Im}(T) \subseteq \mathrm{Ker}(T)$.

b. Let $V = \mathbb{R}^3$ and let $T : V \to V$ be given by seding $\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ x \\ 0 \end{pmatrix}$. Show that $T^2 = 0$ and that $\mathrm{Ker}(T)$ is not contained in $\mathrm{Im}(T)$.

c. Find a V and T with $T^2 = 0$ such that $\mathrm{Ker}(T) \subseteq \mathrm{Im}(T)$.

d. Show that the rank of T is at most $n/2$.

e. On the next page is an attempt by an AI (ChatGPT) to prove that $\mathrm{Ker}(T) \subseteq \mathrm{Im}(T)$. Find the mistake or gap in the proof and explain why it is wrong.

Let V be an n-dimensional vector space and let $T : V \to V$ be a linear transformation such that $T^2 = 0$. This means that for any vector $v \in V$, we have $T(T(v)) = 0$.

Step 1: Analyze the kernel and image: The kernel of T, denoted as $\ker(T)$, is defined as:

$$\ker(T) = \{v \in V \mid T(v) = 0\}.$$

The image of T, denoted as $\text{im}(T)$, is defined as:

$$\text{im}(T) = \{T(v) \mid v \in V\}.$$

Step 2: Show that $\ker(T) \subseteq \text{im}(T)$: To show that $\ker(T) \subseteq \text{im}(T)$, we need to demonstrate that any vector $v \in \ker(T)$ can be expressed as $T(w)$ for some $w \in V$.

1. **Take any $v \in \ker(T)$:** By the definition of the kernel, we know $T(v) = 0$.

2. **Consider $T(v)$:** Since $T^2 = 0$, we have:

$$T(T(v)) = T(0) = 0.$$

This means that $T(v)$ is indeed in the kernel of T.

3. **Construct w:** We want to express v as $T(w)$ for some w. Since $T^2 = 0$, we know $T(w)$ must yield a result that lies in the kernel of T. If we take any $u \in V$, we have $T(u) \in \text{im}(T)$.

Step 3: Relate dimensions: Since $T^2 = 0$, we have $\text{im}(T) \subseteq \ker(T)$. This means every element in the image of T maps to zero under T.

From the rank-nullity theorem, we have:

$$\dim(V) = \dim(\ker(T)) + \dim(\text{im}(T)).$$

Let:

$$k = \dim(\ker(T)) \quad \text{and} \quad m = \dim(\text{im}(T)).$$

Then,

$$n = k + m.$$

Since $\text{im}(T) \subseteq \ker(T)$, it follows that:

$$\dim(\text{im}(T)) \leq \dim(\ker(T)).$$

This implies:

$$m \leq k.$$

Conclusion : If m (dimension of the image) is less than or equal to k (dimension of the kernel), and since $\text{im}(T)$ is a subspace of $\ker(T)$, we conclude that every element of $\ker(T)$ can be expressed in terms of elements in $\text{im}(T)$. Therefore, we find:

$$\ker(T) \subseteq \text{im}(T).$$

Thus, we have proven that the kernel of T is contained in the image of T when $T^2 = 0$.

[Total: 50 marks]